

Improved Algorithms for Contextual Dynamic Pricing

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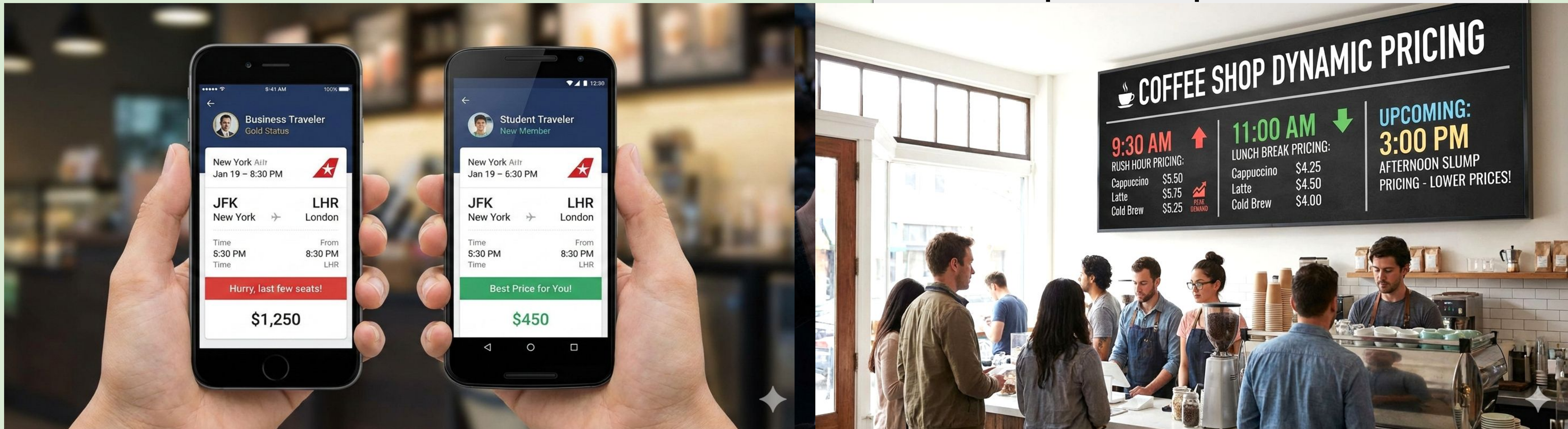
Tullii, Matilde, et al. "Improved algorithms for contextual dynamic pricing." *Advances in Neural Information Processing Systems* 37 (2024): 126088-126117.

Overview

- (Contextual) Dynamic Pricing
- The Mathematical Framework
- Learning Algorithm (Value Approximation - Policy Elimination — VAPE)
- A Quick Demonstration for Toy Example
- Summary and Future Directions

Motivation: What is Dynamic Pricing and Why?

Adjusting prices over time
Seller updates prices based on



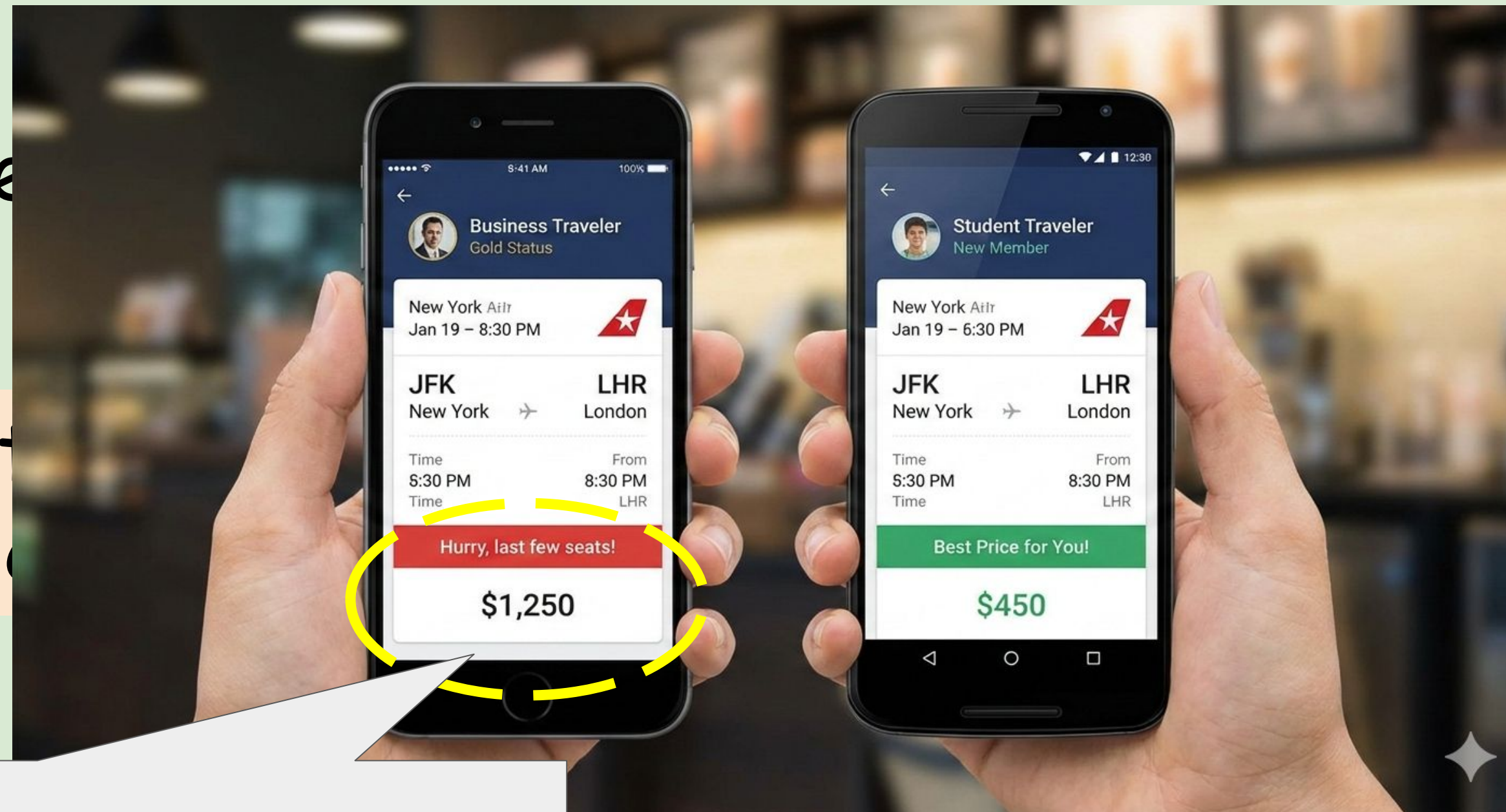
- Customers are different
- Demand changes over time

Goal: Not to just sell more but **balance price and probability of sale** by avoiding underpricing or overpricing to **maximize long-term revenue**.

How to set prices to maximize long-term revenue?

- Customers have different valuations of the product they are purchasing

At any time t
- Loss of chance



product they

purchase

Ticket price $>$ personal valuation
No purchase
Loss of customer
Ticket price $<$ personal valuation
Purchase happens
However, loss of revenue

Goal of seller: Optimally set prices such that they gain profit while satisfying the customer

How? Seller do not know customer personal valuation of the product

Contextual Learning!

Contextual Dynamic Pricing

Customer

Based on context, each customer has a personal valuation (unknown to seller)

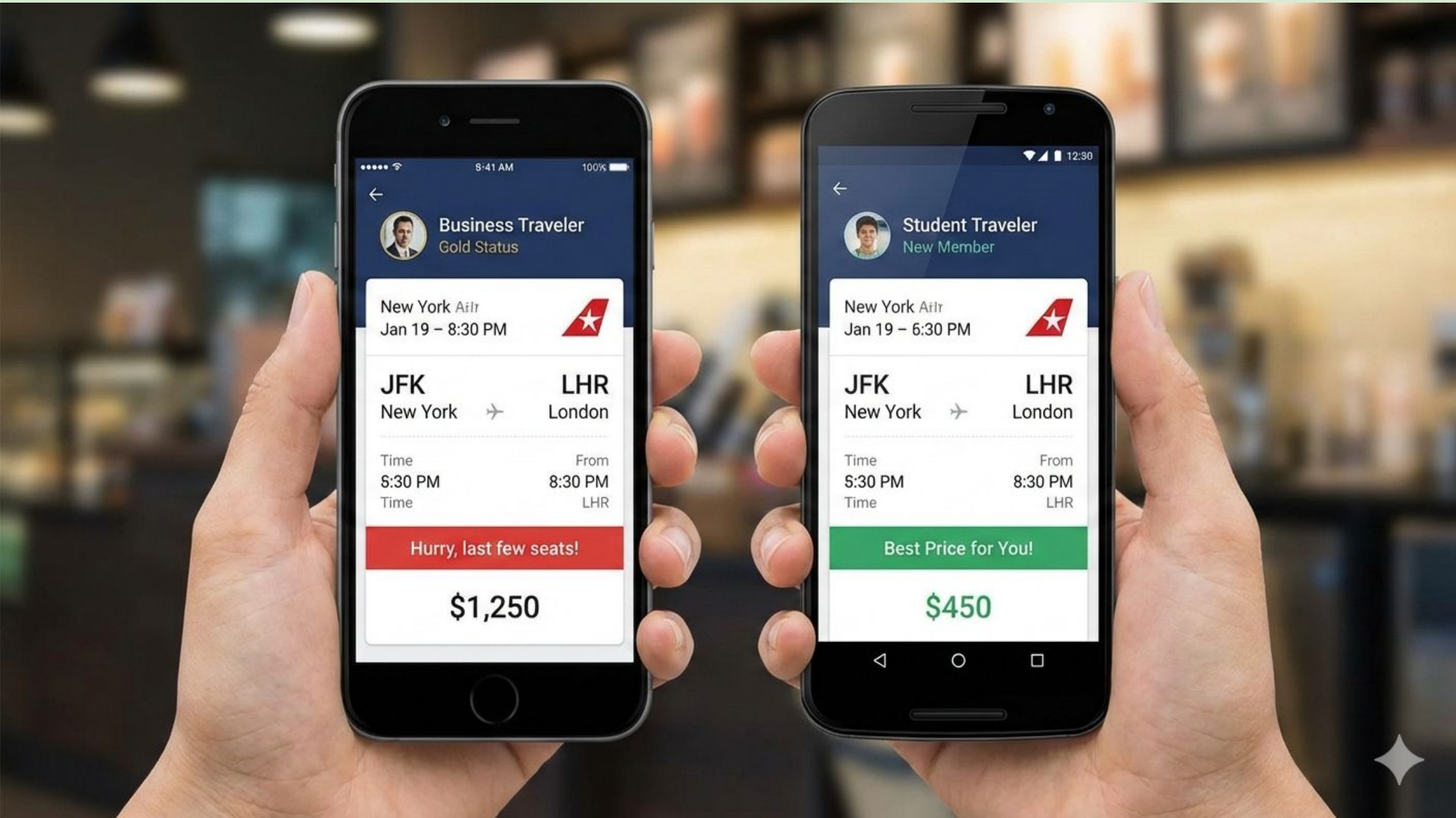
Context
(time,
demand)

Seller **learns and predicts** the valuation of the customer (given context)

Sets the optimal price to make positive revenue, and w.h.p. customer makes a purchase

Seller Model

Contextual Dynamic Pricing - Example



Student

\$400

\$450



Tech Lead

\$800

\$1250



Senior Citizen

\$700

\$725



Seller final price

**Customer
Context (Known
to Seller)**

**Customer
Valuation
(unknown to
seller)**

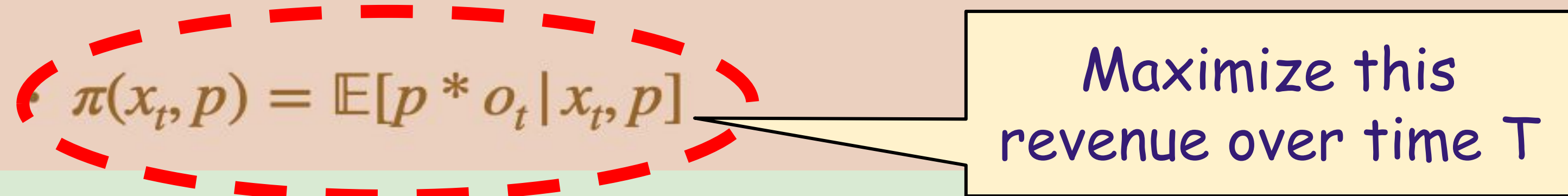
Goal of seller: Optimally set prices to maximize long-term revenue

Contextual Dynamic Pricing

The Mathematical Framework

- At each time $t \leq T$, let the **context** be $x_t \in \mathbb{R}^d$ (known to the seller)
- The **customer** has a **hidden valuation** for the product given by y_t (unknown to the seller)
- The **seller** sets a **price** p_t
- A sale happens if $p_t \leq y_t$. The seller only observe the outcome $o_t = \mathbb{I}\{p_t \leq y_t\}$

- If the **seller's unknown valuation** follows a model: $y_t = g(x_t) + \xi_t$,
 - where $g : \mathbb{R}^d \rightarrow \mathbb{R}$ is the valuation function, and ξ_t is bounded and i.i.d. noise term
- Then, the **total revenue** of the seller if they post price p for the product is given by


$$\pi(x_t, p) = \mathbb{E}[p * o_t | x_t, p]$$

Maximize this
revenue over time T

Regret — For Performance Analysis

- The performance of online algorithms are measured with metric '**regret**'
- Regret is the **difference between the total revenue of the proposed algorithm, and the best offline policy** (suppose the seller knows the customer valuations)

- Here,

$$R_T = \sum_{t=1}^T \max_{p \in \mathbb{R}} \pi(x_t, p) - \sum_{t=1}^T \pi(x_t, p)$$

Revenue of offline
policy at time 't'

- Regret should not grow linear in 't'. Algorithm with sub-linear regret is considered to be a 'good' online-learning algorithm

A word on Naïve Bandits and Regret Bounds

- In contextual dynamic pricing, each context have different valuations, and the optimal prices for each context could be different
- There is no single optimum value
- Naive bandits are good to find single optimum. We have to find optima (vector of optimum values) for all contexts.
- Hence, the regret with naive bandits would be more and is proven to be of order $\tilde{O}(T^{3/4})$

Proposed Improved Algorithm

Value Approximation - Price Elimination(VAPE)

1. Value Approximation
2. Price Elimination

Requirements

1. Noise is i.i.d.
2. Context and Expected Valuation are bounded

Value Approximation Price Elimination(VAPE)

The Idea

Divide the problem into two-parts



- Seller Price = customer valuation + markup
- Phase-1 (value approximation): learn customer valuation in each context
- Phase-2 (price elimination): Based on learned valuation, predict what might be good markup

Pseudocode for VAPE

$$o_t = 1\{p_t \leq y_t\}$$

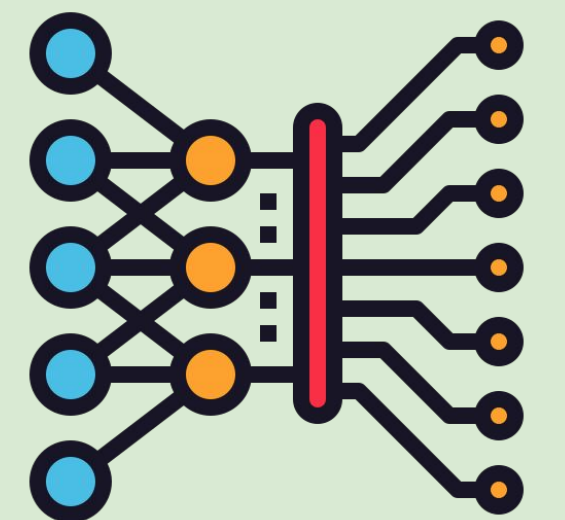
Algorithm 1 VALUATION APPROXIMATION - PRICE ELIMINATION (VAPE): General scheme

- 1: **Input:** Price increments $\{\delta_k\}_{k \in \mathcal{K}}$, expected valuation precision $\text{err}_t(x)$, reward confidence intervals $[\text{LCB}_t(k), \text{UCB}_t(k)]$, parameters α, ϵ .
 - 2: **while** $t \leq T$ **do**
 - 3: **if** $\text{err}_t(x_t) > \epsilon$ **then** ▷ Valuation Approximation
 - 4: Post a price $p_t \sim \mathcal{U}([-B_y, B_y])$
 - 5: Use o_t to improve the valuation estimator $\hat{g}(x_t)$
 - 6: **else** ▷ Price Elimination
 - 7: $\mathcal{A}_t \leftarrow \{k \in \mathcal{K} : \hat{g}_t + \delta_k \in [0, B_y]\}$
 - 8: $\mathcal{K}_t \leftarrow \{k \in \mathcal{A}_t : \text{UCB}_t(k) \geq \max_{k' \in \mathcal{A}_t} \text{LCB}_t(k')\}$
 - 9: Choose $k_t \in \arg \min_{k \in \mathcal{K}_t} N_t^k$ and post a price $p_t = \hat{g}_t + \delta_{k_t}$
 - 10: Update $\hat{D}_{t+1}^{k_t}, N_{t+1}^{k_t}$
-

PHASE 1 - Value Approximation for VAPE

Instead of treating every context as a brand-new problem, VAPE uses a global estimator to estimate the underlying parameters (cross-learning)

$$\tilde{O}(T^{3/4})$$



PHASE 2 - Price Elimination (Bandit) for VAPE

1. The algorithm maintains a set of prices from phase
2. As the algorithm runs, it eliminates prices that are statistically unlikely by shrinking the LCB
3. More efficient than standard ETC because the "elimination" happens dynamically



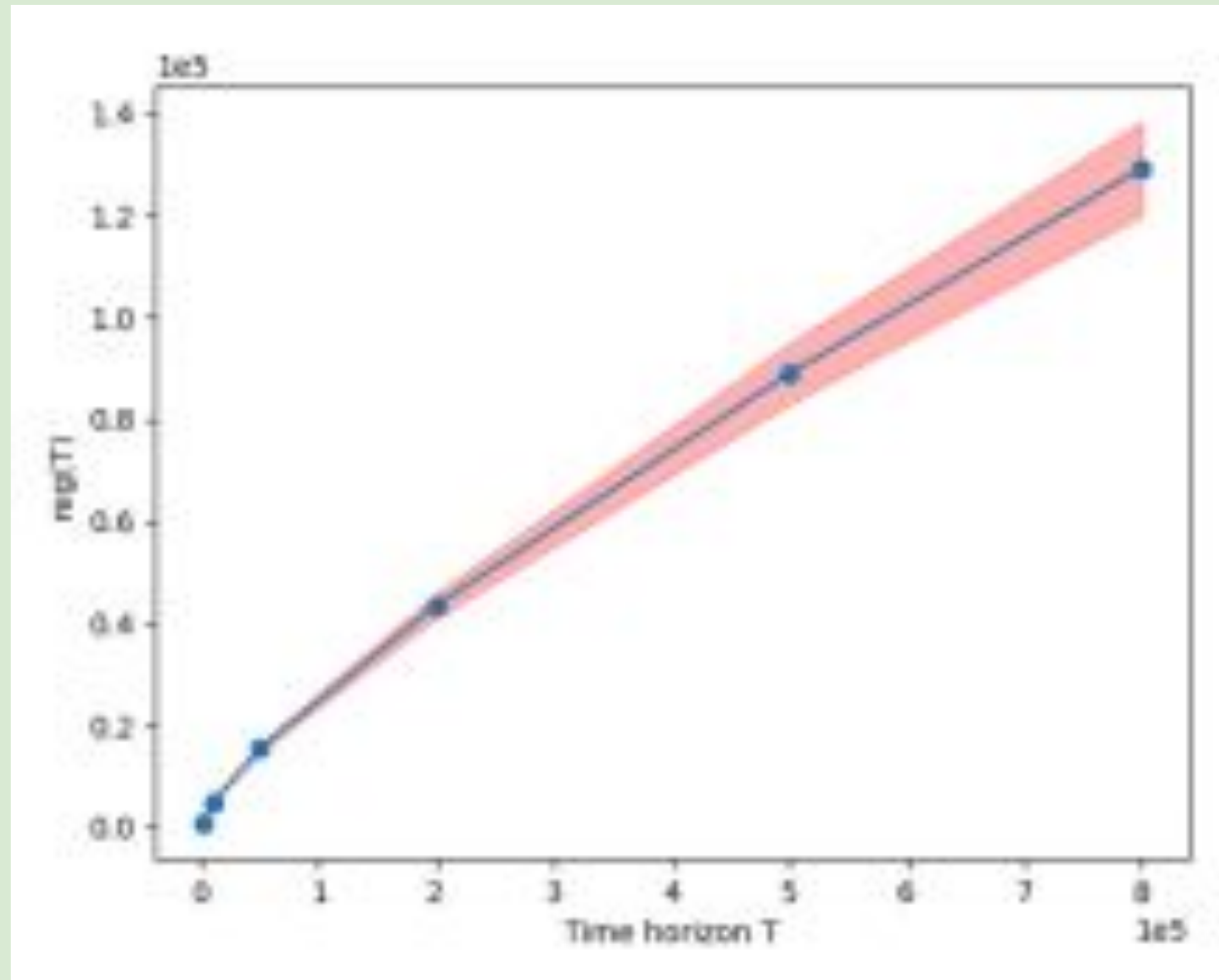
VAPE - Linear Model

1. Noise is bounded and it's c.d.f. is Lipschitz continuous

Valuation equation:

$$g(x) = x^T \theta$$

Regret Graph



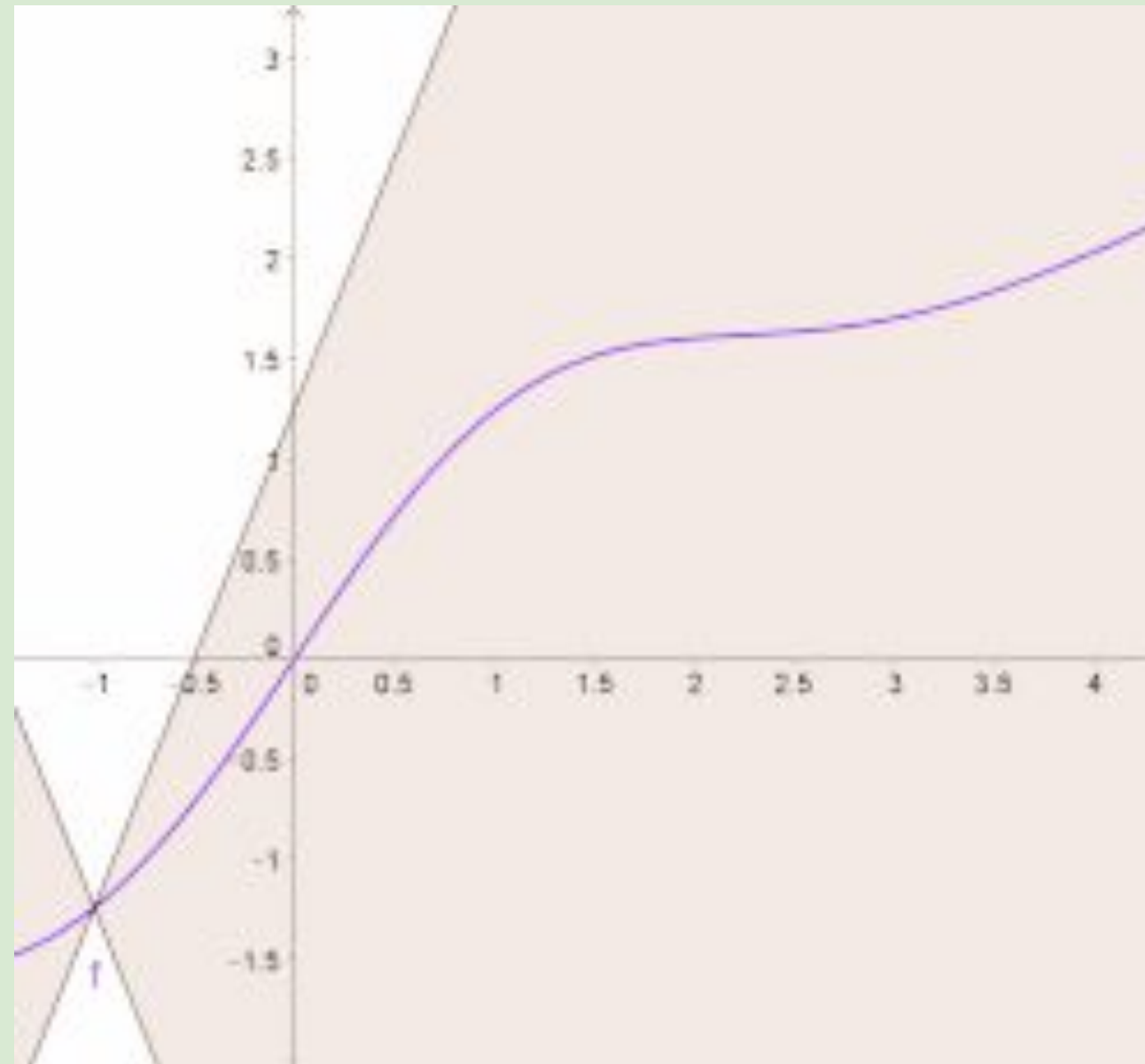
VAPE - Linear Model

Algorithm 2 VALUATION APPROXIMATION - PRICE ELIMINATION (VAPE) for Linear Valuations

- 1: **Input:** bounds B_y and L_ξ , parameters α, μ, ϵ .
 - 2: **Initialize:** $\hat{\theta}_1 = \mathbf{0}_d$, $\mathbf{V}_1 = \mathbf{I}_d$, $K = \lceil (B_y + 1)/\epsilon \rceil$, $\mathcal{K} = \llbracket -K, K \rrbracket$, and for $k \in \mathcal{K}$, $N_1^k = \hat{D}_1^k = 0$.
 - 3: **while** $t \leq T$ **do**
 - 4: **if** $\|x_t\|_{\mathbf{V}_t^{-1}} > \mu$ **then** ▷ Valuation Approximation
 - 5: Post a price $p_t \sim \mathcal{U}([-B_y, B_y])$
 - 6: $\iota_t \leftarrow 1$, $\mathbf{V}_{t+1} \leftarrow \sum_{s \leq t} \iota_s x_s x_s^\top + \mathbf{I}_d$, $\hat{\theta}_{t+1} \leftarrow 2B_y \mathbf{V}_{t+1}^{-1} \sum_{s \leq t} \iota_s (o_s - \frac{1}{2}) x_s$
 - 7: **else** ▷ Price Elimination
 - 8: $\iota_t \leftarrow 0$, $\hat{g}_t \leftarrow x_t^\top \hat{\theta}_t$, $\mathcal{A}_t \leftarrow \{k \in \mathcal{K} : \hat{g}_t + k\epsilon \in [0, B_y]\}$
 - 9: **for** $k \in \mathcal{A}_t$ **do**
 - 10: $\text{UCB}_t(k) \leftarrow (\hat{g}_t + k\epsilon) (\hat{D}_t^k + \sqrt{\frac{2 \log(1/\alpha)}{N_t^k}} + 2L_\xi \epsilon)$
 - 11: $\text{LCB}_t(k) \leftarrow (\hat{g}_t + k\epsilon) (\hat{D}_t^k - \sqrt{\frac{2 \log(1/\alpha)}{N_t^k}} - 2L_\xi \epsilon)$
 - 12: $\mathcal{K}_t \leftarrow \{k \in \mathcal{A}_t : \text{UCB}_t(k) \geq \max_{k' \in \mathcal{A}_t} \text{LCB}_t(k')\}$
 - 13: Choose $k_t \in \arg \min_{k \in \mathcal{K}_t} N_t^k$ and post a price $p_t = \hat{g}_t + k_t \epsilon$
 - 14: Update $\hat{D}_{t+1}^{k_t} \leftarrow \frac{N_t^{k_t} \hat{D}_t^{k_t} + o_t}{N_t^{k_t} + 1}$, $N_{t+1}^{k_t} \leftarrow N_t^{k_t} + 1$.
-

VAPE - Non-Parametric Model

1. Noise is bounded and it's c.d.f. is Lipschitz continuous
2. Valuation function is β -Hölder Continuous



Hölder Continuous

$$|f(x) - f(y)| \leq C|x - y|^\alpha, 1 \geq \alpha > 0$$

VAPE - Non-Parametric Model

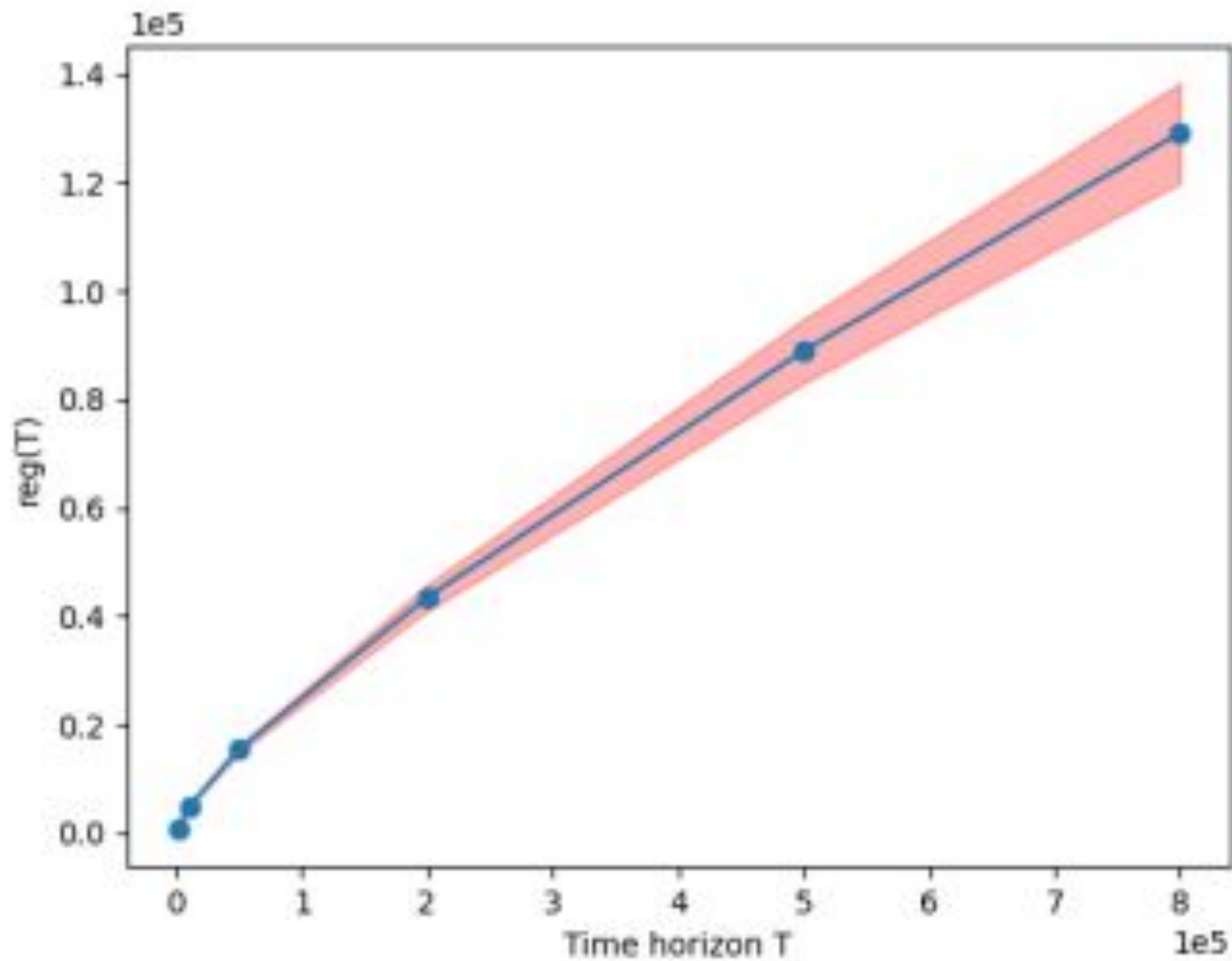
Algorithm 3 VALUATION APPROXIMATION - PRICE ELIMINATION (VAPE) for Non-Parametric Valuations

- 1: **Input:** bounds B_y and L_ξ , finite set $\bar{\mathcal{X}} \subset \mathbb{R}^d$, parameters α, τ, ϵ .
 - 2: **Initialize:** $\mathcal{G}_{\bar{x}} = \emptyset$ for all $\bar{x} \in \bar{\mathcal{X}}$, $K = \lceil B_y + 1/\epsilon \rceil$, $\mathcal{K} = \llbracket -K, K \rrbracket$, and for $k \in \mathcal{K}$, $N_1^k = \hat{D}_1^k = 0$.
 - 3: **while** $t \leq T$ **do** $\bar{x}_t \leftarrow \arg \min_{\bar{x}' \in \bar{\mathcal{X}}} \|x_t - \bar{x}'\|$
 - 4: **if** $|\mathcal{G}_{\bar{x}_t}| < \tau$ **then** ▷ Price Elimination
 - 5: Post a price $p_t \sim \mathcal{U}([-B_y, B_y])$
 - 6: $\mathcal{G}_{\bar{x}_t} \leftarrow \mathcal{G}_{\bar{x}_t} \cup \{t\}$, $\hat{g}(\bar{x}_t) \leftarrow \frac{2B_y}{|\mathcal{G}_{\bar{x}_t}|} \sum_{s \in \mathcal{G}_{\bar{x}_t}} (o_s - \frac{1}{2})$
 - 7: **else** ▷ Run Successive Elimination
 - 8: $\hat{g}_t \leftarrow \hat{g}(\bar{x}_t)$, $\mathcal{A}_t \leftarrow \{k \in \mathcal{K} : \hat{g}_t + k\epsilon \in [0, B_y]\}$
 - 9: **for** $k \in \mathcal{A}_t$ **do**
 - 10: $\text{UCB}_t(k) \leftarrow (\hat{g}_t + k\epsilon) (\hat{D}_t^k + \sqrt{\frac{2 \log(1/\alpha)}{N_t^k}} + 2L_\xi \epsilon)$
 - 11: $\text{LCB}_t(k) \leftarrow (\hat{g}_t + k\epsilon) (\hat{D}_t^k - \sqrt{\frac{2 \log(1/\alpha)}{N_t^k}} - 2L_\xi \epsilon)$
 - 12: $\mathcal{K}_t \leftarrow \{k \in \mathcal{A}_t : \text{UCB}_t(k) \geq \max_{k' \in \mathcal{A}_t} \text{LCB}_t(k')\}$
 - 13: Choose $k_t \in \arg \min_{k \in \mathcal{K}_t} N_t^k$ and post a price $p_t = \hat{g}_t + k_t \epsilon$
 - 14: Update $\hat{D}_{t+1}^{k_t} \leftarrow \frac{N_t^{k_t} \hat{D}_t^{k_t} + o_t}{N_t^{k_t} + 1}$, $N_{t+1}^{k_t} \leftarrow N_t^{k_t} + 1$.
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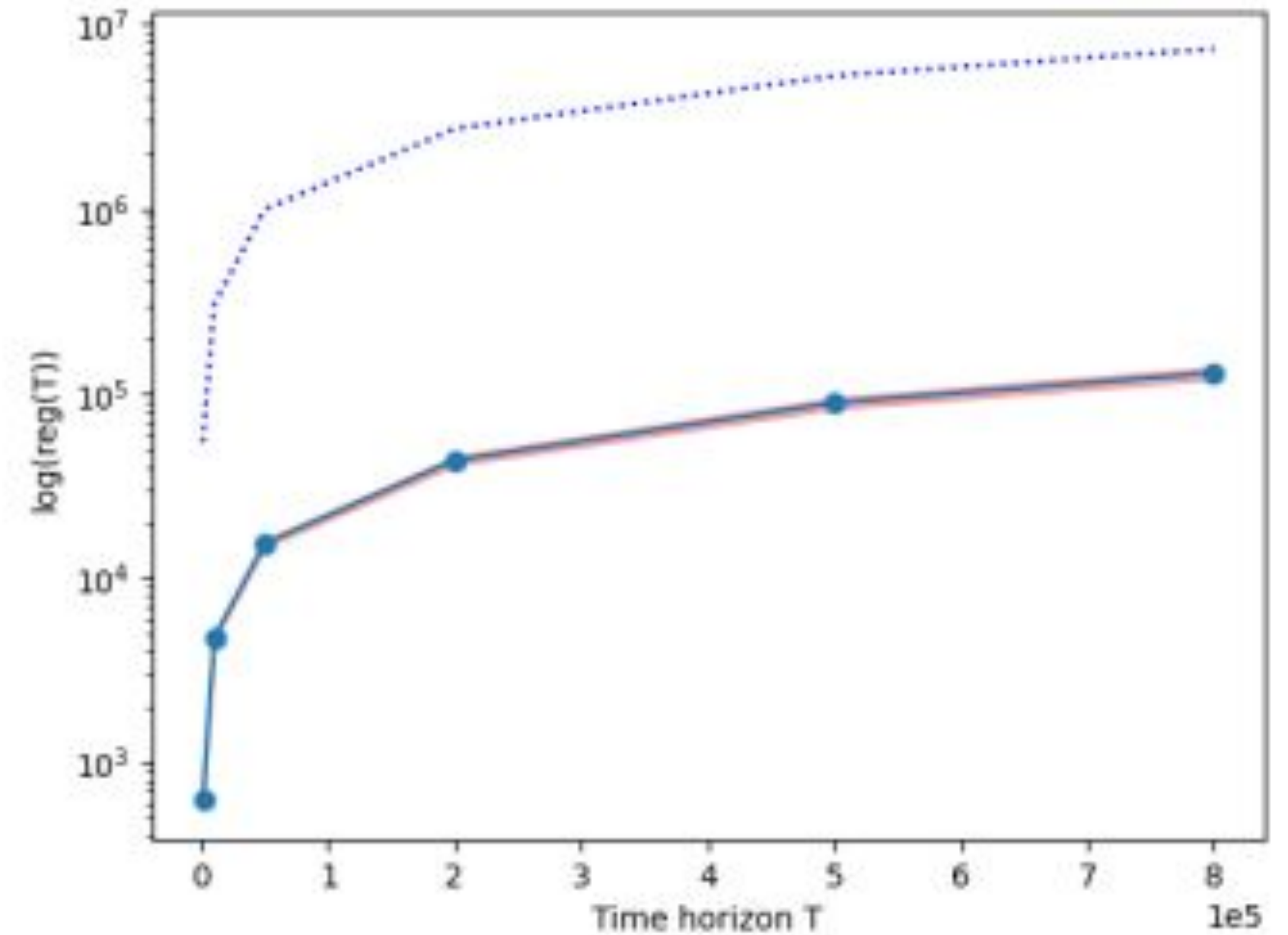
Regret Bounds - The Theoretical Results

Model Type	Regret Bound	Significance
Linear	$\tilde{O}(T^{2/3})$	Matches the theoretical lower bound; it's the "perfect" rate.
Non-Parametric	$\tilde{O}(T^{\frac{d+2\beta}{d+3\beta}})$ Optimal $\tilde{O}(T^{\frac{d+\beta}{d+2\beta}})$	The first rate of its kind with such minimal assumptions (β – Hölder)

Empirical Analysis - Regret Rate

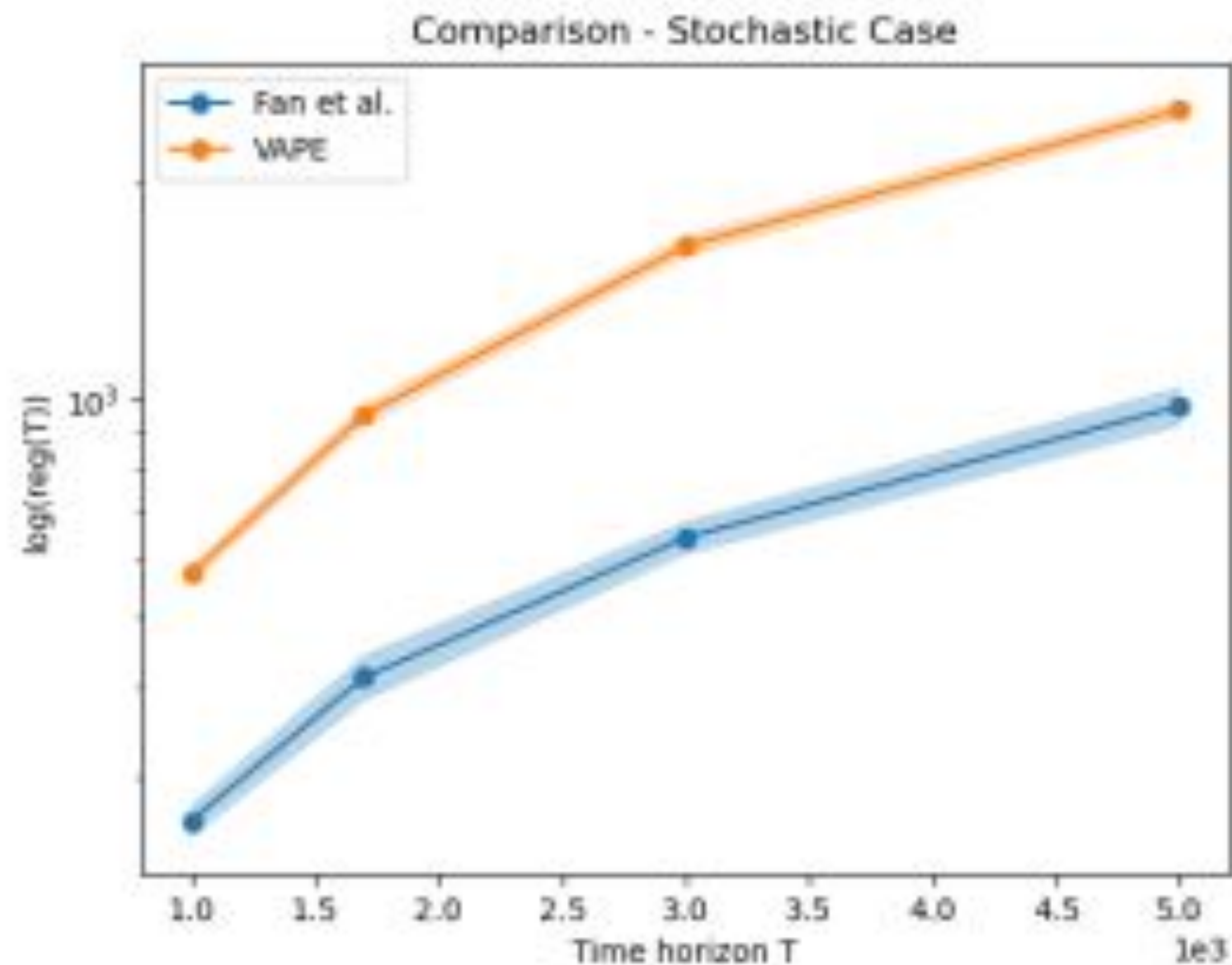


Standard Scale

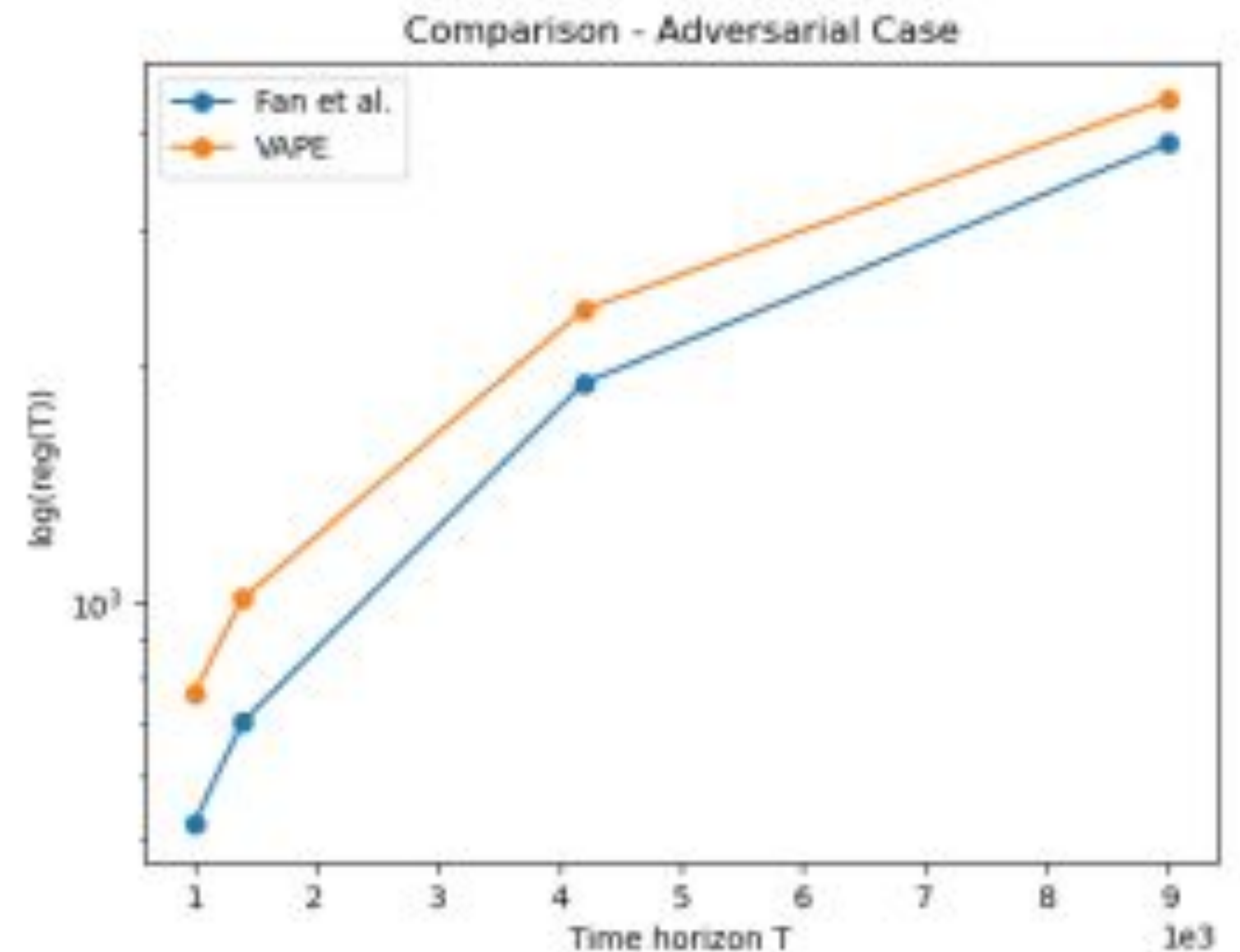


Logarithmic Scale

Empirical Analysis - Regret Rate



$T \in [1000, 1700, 3000, 5000]$



$T \in [1000, 1400, 4200, 9000]$

Demonstration of VAPE for a Toy Example

Customer valuations
based on context
(unknown)

Student	\$400	\$390
Tech Lead	\$800	\$810
Senior Citizen	\$700	\$650

Learned Customer
valuations (VA)-step1

Information-sharing across contexts

Learned markup
(PE) - step2

20\$ markup
lead to best
profit

\$410	✓
\$830	✓
\$670	✓

Final Prices

Step-1 (VA)
Learn expected valuation in
each different context by
exploring several customers

Step-2 (PE)
Explore markup prices that
lead to most sales among all
users (test with different
markup values)

Summary

- Studied the problem of dynamic pricing with contexts
- Studied an online learning algorithm — VAPE, and its applications to linear and non-parametric valuations
- Seen that results improved or maintained (compared to prior studies) despite relaxation of assumptions

Limitations of the work

- Unknown optimal dependence of lower bound $\Omega(T^{2/3})$ on dimension of context
- No similar lower bound for non-parametric
- Noise is assumed i.i.d.

Potential Future Directions

- Evaluate tightness of results
- Explore other valuation models with VAPE
- Further relax assumptions (especially Assumption 2)
- Evaluate when noise is variable

References

1. Gaucher, S., Merlis, N., Perchet, V., & Tullii, M. (2024). Improved algorithms for Contextual Dynamic Pricing. *Advances in Neural Information Processing Systems* 37, 126088-126117. <https://doi.org/10.52202/079017-4006>
2. Wikimedia Foundation. (2025, December 1). *Lipschitz continuity*. Wikipedia. https://en.wikipedia.org/wiki/Lipschitz_continuity#References

Thank You!